Euler's Totient Function and Fermat's Theorem

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Euler's Totient Function

Definition: The totient function $\phi(n)$ counts integers from 1 to n that are relatively prime to n.

Formula:

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right) \tag{1}$$

where p are the distinct prime factors of n.

Fermat's Theorem

Statement: For any integer a such that $\gcd(a, n) = 1$,

$$a^{\phi(n)} \equiv 1 \mod n \tag{2}$$

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Example: Let $n = 9$, then $\phi(9) = 6$

$$2^6 = 64$$

$$64 \mod 9 = 1$$

Thus, Fermat's theorem holds.

Corollary: Euler's Totient Function for Prime $n = p$

If $n = p$ is the prime, then:

$$\phi(n) = n - 1 \tag{3}$$

Example: $n = 7$

$$\phi(7) = 7 - 1 = 6$$

Applying Fermat's theorem:

$$3^6 \equiv 1 \mod 7$$

Corollary: Euler's Totient Function for Composite $n = p.q$

If $n = p \cdot q$ (product of two primes), then:

$$\phi(n) = (p-1)(q-1) \tag{4}$$

Example: $n = 15$, with $p = 3, q = 5$:

$$\phi(15) = (3-1)(5-1) = 8$$
$$2^8 = 256$$
$$256 \mod 15 = 1$$

Summary

- Euler's Totient Function counts numbers coprime to n.
- Fermat's Theorem states that for a, $a^{\phi(n)} \equiv 1 \mod n$.
- For prime n, $\phi(n) = n - 1$.
- For composite $n = p \cdot q$, $\phi(n) = (p-1)(q-1)$.